

RESEARCH PAPER EXPLANATION

Dark Patterns Post-GDPR

Scraping Consent Interface Designs and Demonstrating Their Influence

A concise guide to the paper's problem, importance, findings, research questions, methodology, results and discussion

Core issue: Cookie banners may record “consent” without reflecting a genuinely free, informed or privacy-preserving choice.

Prepared from the uploaded research paper by Nouwens et al. (CHI 2020)

Quick Overview

The paper examines whether web consent interfaces actually protect users or instead manipulate them into accepting tracking. The researchers combine a large-scale website analysis with a controlled user experiment. Their central finding is that interface design materially changes consent rates, meaning that recorded consent often reflects design pressure rather than genuine privacy preference.

Guiding Question	Answer
What problem are they solving?	They investigate whether cookie banners obtain real, freely chosen consent or merely create the appearance of consent while nudging users toward tracking.
Why is it important?	Personal data may be shared with hundreds of third parties. If the interface is manipulative, acceptance may be legally recorded but ethically weak and potentially non-compliant.
What did they find?	Dark patterns are widespread, minimum compliance is low, and small design changes - especially hiding "Reject All" - substantially increase acceptance.

1. Title

Meaning of the title

"Dark patterns" are interface techniques that steer users toward choices that benefit the company. "Scraping consent interface designs" refers to the automated collection and analysis of cookie-banner implementations. "Demonstrating their influence" means experimentally testing whether those designs change users' decisions.

Main interpretation

The title signals that the paper is not only describing consent banners; it is testing whether post-GDPR design practices undermine user autonomy and the legal quality of consent.

2. Abstract

What the abstract covers

The abstract summarizes two linked studies: a scrape of 680 consent-management interfaces from leading UK websites and a field experiment with 40 participants exposed to eight consent designs.

Key findings

Only 11.7% of interfaces met the researchers' minimum GDPR-based criteria. Removing "Reject All" from the first page increased consent by about 22-23 percentage points. Showing more granular controls on the first page decreased consent by about 8-20 percentage points. Banner versus barrier style did not significantly change the acceptance rate.

3. Introduction: First 1-2 Paragraphs

Starting point

The paper begins with the “notice and consent” model: users are shown information and are expected to accept, reject or adjust data-processing choices.

Underlying problem

Users usually visit a website to complete another task, not to study privacy settings. Consent banners interrupt that task. Because the website controls the design, it can make accepting quick and rejecting difficult.

Legal benchmark

Under the GDPR, consent must be freely given, specific, informed and unambiguous, and it must result from a clear affirmative action. Scrolling, continuing to browse or closing a notice should not automatically count as valid consent.

4. Conclusion

Overall conclusion

Illegal or highly questionable consent practices remain widespread after the GDPR. Information and controls hidden behind extra pages are rarely used, so they provide weak practical control.

Recommended direction

The authors call for stronger enforcement, automated auditing, responsibility for CMP providers, and more durable browser-level privacy controls rather than repeated site-by-site consent decisions.

Broader meaning

The current model gives companies control over the decision environment while placing the burden on users. A recorded click is therefore not enough to prove that consent was meaningful.

Applying the Paper to Five Broader Questions

1. How much user data should a web application collect?

Only data that is necessary, relevant and proportionate to a clearly defined purpose. Essential data for login, security or a shopping cart is different from optional profiling, advertising and cross-site tracking. Optional collection should be off by default and retained only as long as needed.

2. Is user consent alone sufficient to make data collection ethical?

No. Consent is ethically weak when rejection is difficult, choices are preselected, information is hidden, users cannot understand the consequences, or access is unfairly conditioned on acceptance. Ethical collection also requires necessity, fairness, transparency, security, purpose limitation and accountability.

3. How do dark patterns influence users' privacy decisions?

They exploit convenience, impatience, fear of lost functionality and consent fatigue. The clearest result is that hiding "Reject All" on the first page increased acceptance by 22-23 percentage points, even though users' underlying privacy values may not have changed.

4. What responsibilities do software engineers have?

Engineers shape autonomy through architecture and interface decisions. They should apply privacy-by-design, data minimisation, equal prominence for acceptance and rejection, no preselected optional tracking, no processing before valid consent, easy withdrawal, truthful consent logging and escalation of manipulative product requirements.

5. How can companies balance personalization and privacy?

Offer a useful basic service without unnecessary tracking. Request optional personalization separately, explain the concrete benefit, collect the minimum data required, prefer first-party or on-device processing, and let users revise or withdraw preferences easily. Personalization should be a transparent value exchange, not disguised coercion.

5. Research Questions

The paper asks two official questions:

- What is the current state of consent-management-platform interface design in the EU, and how prevalent are non-compliant design elements?
- How do interface designs affect users' consent actions and, by extension, whether consent is genuinely freely given?

Research Question	What it examines
RQ1	The real-world prevalence of implied consent, hidden rejection, preselected settings and other potentially non-compliant design choices.
RQ2	Whether users' decisions change when interface structure, button prominence and control granularity change.

6. Methodology

Study 1: Website scraping

The researchers built a web scraper using Scrapy and Splash. It examined the top 10,000 UK websites and successfully collected 680 consent interfaces from five major CMP providers: QuantCast, OneTrust, TrustArc, Cookiebot and Crownpeak.

They assessed three minimum conditions:

- Consent must be explicit.
- Rejecting all must be as easy as accepting all.
- Optional purposes and vendors must not be preselected.

Study 2: User experiment

Forty participants installed a Chrome extension that injected eight different consent-interface designs during normal browsing. The designs varied by notification style, availability of accept/reject buttons and the granularity of purposes or vendors shown on the first page. The researchers recorded interactions and used regression analysis to estimate the effect of each design.

Component	Sample	Main Variables	Purpose
Study 1	680 CMP interfaces	Explicit/implicit consent; button access; preselection	Measure real-world design and minimum compliance
Study 2	40 participants; 8 designs	Banner/barrier; button prominence; granularity	Measure causal influence on consent behavior

7. Results

Website analysis

Finding	Meaning
Implicit consent	32.5% of sites used some form of implied consent.
Reject All missing	50.1% did not provide a "Reject All" button.
Equal ease	Only 12.6% made rejecting as easy as accepting.
Preselected options	56.2% preselected optional vendors or purposes.
Minimum compliance	Only 11.7% met all three minimum conditions.
Third parties	Sites listing vendors had a median of 315 vendors.

User experiment

- Banner versus barrier did not significantly change acceptance rates.
- Removing "Reject All" increased acceptance by about 22-23 percentage points.
- Showing detailed purposes or vendors reduced acceptance by about 8-20 percentage points.
- 93.1% of interactions remained on the first page.

- Only about 1.3% of all responses represented genuinely personalised selections.
- Many participants later said their actual choices did not match their ideal privacy preferences.

Interpretation: The consent outcome changes when the interface changes. Therefore, the click itself cannot be treated as a pure measure of the user's underlying preference.

8. Discussion

The study shows that consent is not merely an individual decision; it is partly engineered by the decision environment. When rejection is hidden and accepting is immediate, the interface shifts behavior toward the company's preferred outcome.

Most important design insight

Anything hidden beyond the first screen is unlikely to provide meaningful control.

This creates a gap between formal consent and meaningful consent. A user may technically click "Accept," yet the choice may be shaped by friction, fatigue, incomplete information or fear that the service will not work.

Who should carry responsibility?

- Website operators
- Consent-management-platform providers
- Software engineers and UX designers
- Advertising platforms
- Regulators and data-protection authorities

Limitations

The user experiment involved only 40 participants who were mostly young and university-educated. It was conducted in the United States even though the legal analysis concerns Europe. The exact percentages should therefore not be generalized automatically to every population. However, the study still provides strong causal evidence that interface design can materially influence privacy decisions.

Final Takeaway

Valid privacy design is not just about displaying a consent box. It requires minimising data collection, presenting genuine and balanced choices, preventing manipulation, and making refusal as easy as acceptance.

Source paper: M. Nouwens, I. Liccardi, M. Veale, D. Karger and L. Kagal, "Dark Patterns Post-GDPR: Scraping Consent Interface Designs and Demonstrating their Influence," CHI 2020.